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Dynamic Analysis for Employment Substitution Effect of Robots Fused with Deep Learning Models and Artificial Intelligence

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Abstract: The purpose of this study is to address the problem of separation between temporal dynamics and network structure in analyzing the employment substitution effect of robots. The study proposes a dynamically coupled model integrating Graph Convolutional Network (GCN) and Bidirectional Long Short-Term Memory (Bi-LSTM). In the temporal dimension, the framework adopts Bi-LSTM to capture the forward and backward dependencies of technological penetration. Meanwhile, in the structural dimension, this study introduces GCN to model the skill correlation network between jobs. The core innovation lies in designing a gated fusion mechanism, which dynamically integrates temporal features and network embeddings at each time step to achieve a refined simulation of the interaction process. The experiment is based on panel data of Chinese manufacturing enterprises from 2018 to 2022, covering 327 samples. Results show that the proposed model achieves an accuracy of 0.912 and an Area Under the Curve value of 0.928 on the test set, significantly outperforming single Bi-LSTM, GCN, and linear regression models. A key finding is that the model can accurately identify pre-substitution states with a prediction accuracy exceeding 89%, providing a 6-12-month early warning window for policy interventions. Gating analysis reveals that the contribution rates of temporal dynamics and skill networks are 48.3% and 42.7%, respectively, verifying the existence of a dual-drive mechanism. The study confirms that AI substitution is not an immediate event but a gradual process involving multiple stages. This model provides a new paradigm for understanding the interaction between technological change and the labor market, shifting policy formulation from passive response to proactive guidance.

Keywords: Artificial intelligence; Deep learning; Robot employment; Substitution effect; Dynamic analysis

1. Introduction

In 2022, the Chat Generative Pre-trained Transformer (ChatGPT), a pre-trained generative chatbot with human-computer dialogue capabilities, was officially released. The topic related to the substitution effect of artificial intelligence (AI) technology on human labor has once again sparked widespread discussions in both academic and industrial circles. There is a prevailing concern that, in the future, AI will become a substantial factor in replacing human labour, which may result in widespread unemployment [1,2]. Wang et al. (2023) [3] argued that the application of AI algorithms in enterprise resource optimization became an important path to promote the implementation of the United Nations Sustainable Development Goals. Over the past decades, the density of industrial robots has increased in major economies, with China, South Korea, Singapore, and other countries experiencing particularly rapid deployment. Although technological progress may impact employment in the short term, it can create new jobs in the long run. However, AI-driven automation exhibits distinct characteristics: it not only replaces physical labor but also begins to erode the job space of cognitive and procedural tasks [4-6]. This expansion of the capability boundary has enabled the breadth and depth of employment substitution to far exceed those of previous technological revolutions.

The large-scale application of AI, on the one hand, improves enterprises' production efficiency and provides them with opportunities to produce higher-level products [7,8]. On the other hand, the application of AI, especially the use of industrial robots in the manufacturing sector, also brings new challenges to employment in China [9,10]. Automation driven by industrial robots will replace a large number of human jobs on a large scale, leading to mass unemployment in the manufacturing industry. Robot substitution does not proceed at a constant speed, and its impact shows remarkable heterogeneity across different industries and skill levels. Assembly line workers face direct substitution risks, while high-skilled engineers may achieve efficiency improvements through technological collaboration. In addition, traditional econometric models have a limited ability to characterize dynamic processes [11]. This simplified processing model may underestimate or misjudge the real impact.

Deep learning (DL), especially Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) networks, provides new ideas for solving the above problems. Such models are good at processing time-series data and can effectively learn nonlinear dynamics in complex systems. Their core advantage lies in automatic feature extraction—without presetting variable combinations, the models can learn potential patterns from raw data. In this study, the Graph Convolutional Network (GCN) is introduced at the static feature extraction end to further realize gated fusion of the bidirectional-LSTM (Bi-LSTM)'s hidden states and GCN's node embeddings at each time step. Through in-depth dynamic feature characterization and predictive analysis of the mechanism by which industrial intelligence affects employment, it is possible to enrich the empirical research on technological progress and employment. Meanwhile, it can provide a verifiable empirical basis for theoretical analysis in related fields and improve the theoretical system in these fields.

2. Literature Review

The development of AI is hailed as the Fourth Industrial Revolution. Each industrial revolution is accompanied by the demise of some traditional industries, which is the substitution effect of technological progress, and this effect is inevitable. Qin et al. (2024) [12] proposed the task-biased technological progress theory. They argued that automation prioritizes replacing medium-skill, routine tasks, leading to job polarization. High-skill and low-skill jobs grow while the middle layer shrinks. This theory explains the structural changes in the labor markets of developed countries but fails to delve into the micro-mechanisms of technological substitution. Wang & Jiao (2025) [13] believed that AI had strong substitutability and tended to replace complex labor. Such a substitution could bring significant risks to China in the short term, so constant attention must be paid to the unemployment risks of low-skill workers. Lane & Saint-Martin (2021) [14] sorted out the changes in production systems during previous industrial revolutions and found that industrial robots were more capable of meeting the market demand for mass customization under Industry 4.0 conditions. They could change the development model of the manufacturing industry in the future and put forward higher requirements for the improvement of workers' skills.

The work of Karaahmetoğlu et al. (2023) [15] constituted an important turning point. They adopted expert questionnaires and machine learning (ML) classifiers to assess the automation potential of different occupations. A support vector machine (SVM) was used to identify the common characteristics of high-substitution-risk jobs, including repetitiveness, environmental stability, and low social complexity. This model relied on manually labeled feature vectors, such as scores on dimensions like perceptual ability and creativity, resulting in strong subjectivity. Brugere et al. (2023) [16] systematically sorted out the patterns of tasks applicable to ML and proposed that predictability and data richness were the keys determining substitution possibility. They pointed out that the success of deep neural networks in domains such as image recognition and natural language processing (NLP) signified the increasing automation of tasks, including visual inspection and fundamental copywriting. Guo et al. (2025) [17] used an RNN to analyze online recruitment advertisement texts and track the temporal evolution of skill demands. By converting unstructured text into vector sequences through word embedding technology, the model captured the increase in the frequency of certain keywords, indirectly reflecting the skill restructuring pressure brought by technological substitution. In the innovation of spatial-temporal feature fusion mechanisms, most existing studies adopt a serial fusion mode of “spatial first and then temporal” or “temporal first and then spatial”. For instance, spatial features are first extracted via GCN and then fed into temporal models to capture temporal features, or the process is reversed. The drawback of serial fusion lies in the failure to realize real-time interactive fusion of spatial-temporal features; it ignores the mutual influence between spatial structure evolution and temporal feature variation. Thus, it is difficult to describe their dynamic coupling relationship. In addition, some studies attempt to adopt the attention mechanism to achieve parallel fusion of spatial-temporal features. However, the attention mechanism has a large parameter scale; this easily leads to overfitting under small-sample conditions and lacks a physical interpretation of feature fusion.

With the swift socio-economic development and profound changes in people's lifestyles, AI can exert a profound impact on people's thinking modes and behavioral patterns [18-20]. After decades of development, the technologies included in AI mainly cover computer vision, ML, NLP, speech recognition, and other aspects. The corresponding technical applications are exhibited in Table 1.

Table 1: AI-related technologies and applications

Technology name	Concept	Main application
Computer vision	Cameras and computers are used to replace human eyes for target recognition, tracking, and measurement, etc. Through image processing, the computer processes the images into ones that are more suitable for human eyes to observe or send to instruments for detection.	Facial recognition systems, industrial automation inspection, autonomous driving navigation, medical image analysis, intelligent security monitoring, and augmented reality applications
ML	A multidisciplinary field involving probability theory, statistics, approximation theory, etc., studying how computers can simulate human learning behavior to acquire new knowledge or skills, reorganize existing knowledge structures to improve their own performance, which is the core of AI.	Personalized recommendation systems, image recognition and classification, financial risk prediction, autonomous driving decision support, medical diagnostic assistance, and industrial equipment fault prediction
NLP	The technology of using computers and software to acquire the meaning of human language (written or spoken), including natural language generation and natural language understanding, processing, and analyzing text or speech data through DL models.	Machine translation, intelligent customer service chatbots, sentiment analysis, voice assistants, spam filtering, automatic summarization, and intelligent writing assistance
Speech recognition	The technology that converts human speech signals into text or commands, serving as a key input link for NLP, achieves the conversion from speech to text through acoustic and language models.	Voice input methods, voice assistant interaction, smart in-car systems, automatic meeting transcription, voice-controlled smart home, automated telephone customer service systems

Analysis reveals that the models designed by the existing research do not adequately describe the intermediate states of the substitution effect. This study aims to construct a dynamic coupling model integrating multimodal DL. The main body of the model adopts a Bi-LSTM layer to process time series data, enabling a more complete modeling of this complex temporal logic. Through this in-depth dynamic analysis, it is expected to provide a more time-sensitive and structure-insightful decision-making basis for technological governance.

3. Research Methodology

3.1 Architecture of the Bi-LSTM Model

Industrial intelligence includes two core dimensions: automation and intellectualization, and their impact paths on employment are significantly different.

Automation technology mainly replaces routine operational tasks in manufacturing, such as repetitive physical labor in assembly, packaging, and quality inspection [21-23]. Such technologies achieve task standardization through program presets, forming a direct substitution for blue-collar workers performing routine operations. For instance, the popularization of welding robots in the automotive manufacturing industry has markedly reduced the demand for welding workers. The impact mechanism of industrial intelligence on labor employment is illustrated in Figure 1. Considering spatial factors further, industrial intelligence exerts a spatial spillover effect on labor employment.

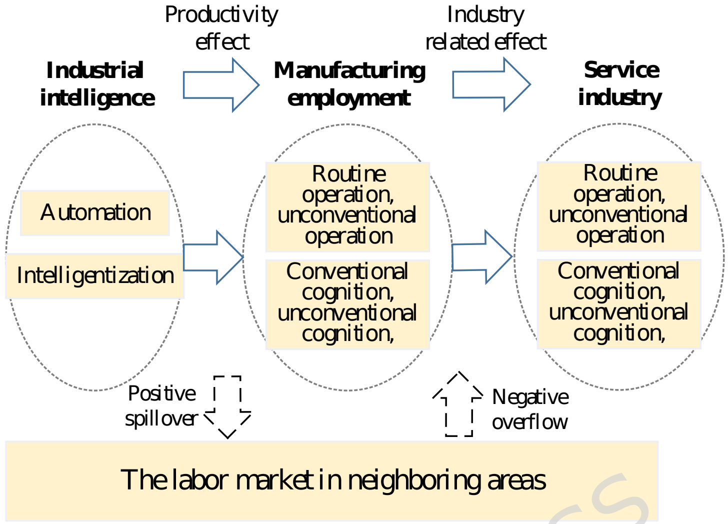

Figure 1: The impact mechanism of industrial intelligence on labor employment

This study adopts Bi-LSTM as the core temporal modeling unit, designed to capture the bidirectional dependencies of contextual information in the employment substitution process. Unlike traditional one-way LSTM, Bi-LSTM achieves comprehensive information capture of time series through two parallel LSTM layers—one processing the sequence from front to back and the other from back to front. This design enables the model to simultaneously utilize contextual information before and after the current moment, providing a more accurate portrayal of the dynamic evolution of the employment substitution effect. The structure of LSTM is revealed in Figure 2. A single LSTM layer comprises one recurrent structure whose self-updating frequency depends on input data dimensions and recursion count. The architecture does not involve multiple interconnected recurrent units within the same layer.

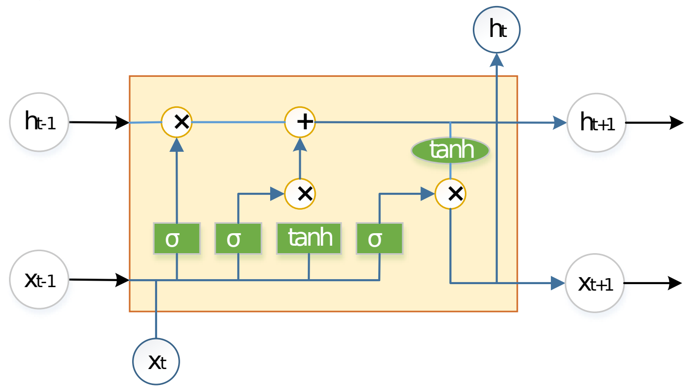

Figure 2: Basic structural units of LSTM

The structure of Bi-LSTM comprises two LSTM layers, processing the forward and backward directions of the input sequence, respectively. The network structure of Bi-LSTM is presented in Figure 3. Each LSTM unit contains three key components: forget, input, and output gates, which precisely control information flow through a gating mechanism. The forget gate determines which historical information to retain, the input gate controls the degree of new information integration, and the output gate decides the hidden state output at the current moment. In employment substitution analysis, these gating mechanisms enable the model to automatically identify key time nodes, such as the launch of robot deployment and the period of skill restructuring. The backward LSTM processes data from the starting point of time, capturing the impact of historical trends on the current state. It can identify how changes in robot density over the past three months affect the current number of jobs. The reverse LSTM traces back from the end of time, analyzing the shaping effect of future expectations on current decisions [24-26].

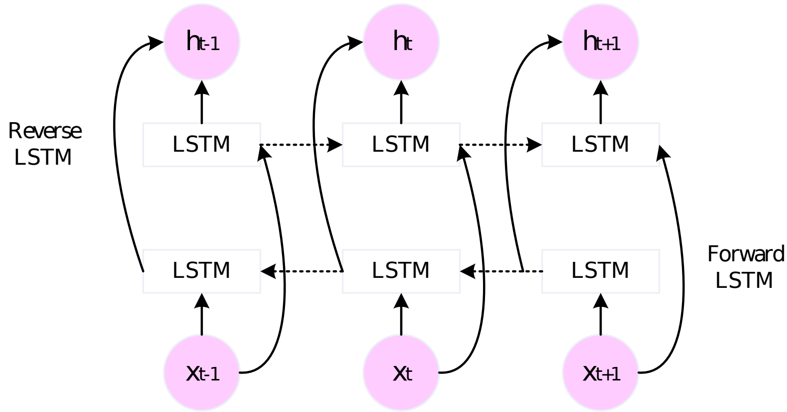

Figure 3: Bi-LSTM network structure

In Figure 3, the forward LSTM structure's calculation process in the BiLSTM network is similar to that of a single LSTM. The BiLSTM's hidden layer state is obtained by combining the forward and backward hidden layer states, and its calculation is:

$$\overrightarrow{h_t} = LSTM(h_{t-1}, x_t) \tag{1}$$

$$\overleftarrow{h_t} = LSTM(h_{t+1}, x_t) \tag{2}$$

$$h_t = a\overrightarrow{h_t} + b\overleftarrow{h_t} \tag{3}$$

x_t refers to the input data at time t; $\overrightarrow{h_t}$, and $\overleftarrow{h_t}$ denote the output of the forward and backward LSTM hidden layers, respectively; a and b stand for constant coefficients, representing the weights.

3.2 Dynamic Analysis Model of Substitution Effect Coupled with GCN

The constructed dynamic analysis model of substitution effect coupled with GCN is a key technical carrier for realizing the dynamic simulation of the employment substitution effect. By deeply integrating temporal dynamics and network structure information, the model achieves a refined characterization of the employment substitution process. GCN is a neural network model specifically designed to process graph-structured data. It can perform convolution operations on graphs to capture relationships and contextual information between nodes. The structure of GCN is displayed in Figure 4. In the process of spatiotemporal feature extraction, the GCN model considers information in both spatial and temporal dimensions simultaneously. The spatial graph convolution operator is defined according to the spatial relationships between a node and its neighboring set. A straightforward computational approach involves averaging the feature vectors of the central node and its adjacent nodes.

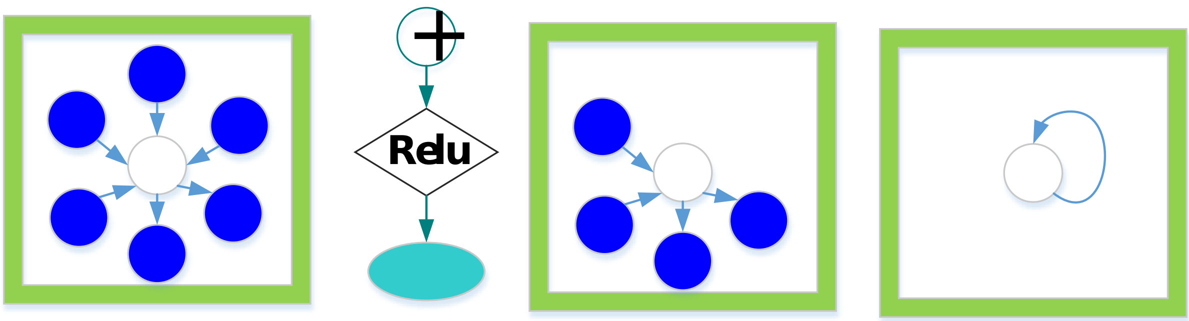

Figure 4: GCN structure

The graph is represented as $G=\{V,E\}$, where V and E refer to the set of nodes and edges. For spatial partitioning, this study divides the neighboring node set into three subsets and uses the centroid of the joint coordinates as the partitioning center. Let the adjacency matrix be denoted as A and the degree matrix as D, it can obtain:

$$D(i,i)=\sum_{j} A(i,j)+\alpha \tag{4}$$

$$A^{norm}=D^{-\frac{1}{2}}AD^{-\frac{1}{2}} \tag{5}$$

$D(i,i)$ is a diagonal matrix, and the final graph convolution operation can be expressed as:

$$f_{out}(v_i)=\sum_{v_j \in B_i} \frac{1}{Z_{ij}} f_{in}(v_j)\cdot w(l_i(v_j)) \tag{6}$$

B_i and $f(*)$ represent a neighboring set and a feature representation of the node, respectively; $w(*)$ is the weight calculation. By sampling, the set of neighboring nodes can be obtained, and then a function is selected to aggregate the features of adjacent nodes within the node's neighborhood.

The above calculation method is relatively complex, so by introducing a parameterized smooth factor, the spectral filter is localized in space. An approximation of $g_{\theta}(\Lambda)$ can be obtained using a truncated expansion of the Chebyshev polynomial $T_k(x)$ up to the k-th order:

$$g_{\theta'}(\Lambda) \approx \sum_{k=0}^{K} \theta'_k T_k(\tilde{\Lambda}) \tag{7}$$

$$\tilde{\Lambda}=\frac{2}{\lambda_{max}}\Lambda - I_N \tag{8}$$

$\tilde{\Lambda}$ represents the eigenvector matrix scaled by the largest eigenvalue of L, that is, the spectral radius. $\theta' \in R^N$ denotes a vector of Chebyshev coefficients, and the Chebyshev polynomials are defined using a recursive equation.

$$T_k(x)=2xT_{k-1}(x)-T_{k-2}(x) \tag{9}$$

$T_0(x)=1$ and $T_1(x)=x$.

After processing through the above steps, the spectral graph convolution no longer depends on the entire graph, but only on the neighboring nodes within K steps from the central node. At this point, spectral graph convolution can be approximated as a linear function of $\tilde{L}$. Further, in the linear model of GCN, by approximating $\lambda_{max} \approx 2$, the first-order linear approximation of this convolution can be obtained:

$$g_{\theta'} * x \approx \theta'_0 x+\theta'_1 (L-I_N)x=\theta'_0 x-\theta'_1 D^{-\frac{1}{2}}AD^{-\frac{1}{2}}x \tag{10}$$

θ'_0,θ'_1 is a free parameter. To avoid overfitting, the parameter is constrained, denoted as $\theta=\theta'_0=-\theta'_1$, resulting in equation (11):

$$g_{\theta'} * x=\theta(I_N+D^{-\frac{1}{2}}AD^{-\frac{1}{2}})x \tag{11}$$

From this, the final form of the GCN can be derived:

$$H^{n+1}=\sigma(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^n W^n) \tag{12}$$

The network's input in the *n*th layer is $H^n \in R^{N*D}$; *N* is the number of nodes in the graph, and each node is represented by a *D*-dimensional feature vector. $\tilde{A} = A + I_N$ represents the adjacency matrix with self-connections added; $\tilde{D} = \sum_j \tilde{A}_{ij}$ denotes the D-degree matrix; $W^N \in R^{N*D}$ is the trainable weight parameters; σ refers to the activation function.

The dynamic coupling layer is the proposed model's core innovation. At each time step, it performs gating fusion of the Bi-LSTM's hidden state and GCN's node embedding. The fusion mechanism dynamically adjusts the two feature types' contribution ratio through learnable gating units. The gating signal is calculated based on the current time step's context information, enabling this model to adaptively determine when to focus on temporal dynamics and when to pay attention to the skill network structure [27-29]. The fusion expression is:

$$h_t = g_t \odot h_t^f + (1 - g_t) \odot h_t^b + \alpha \odot e_t \tag{13}$$

h_t^f and h_t^b are the hidden states of the forward and backward LSTM; e_t denotes the node embedding generated by the GCN; α refers to an additional fusion coefficient. g_t is the gating signal, which can be calculated using the sigmoid function:

$$g_t = \sigma\left(W_g \left[h_t^f, h_t^b, e_t\right] + b_g\right) \tag{14}$$

This design enables the model to automatically learn the relative importance of different information sources. During periods of rapid technological diffusion, the weight of GCN features may markedly increase, reflecting the swift restructuring of skill networks. In periods of technological stability, Bi-LSTM features may dominate, capturing more continuous temporal patterns. The introduction of a gated fusion mechanism allows the model to dynamically respond to the features at various stages of the employment substitution process [30].

The core of GCN is to extend convolution operations from Euclidean space to non-Euclidean graph-structured space. Spectral graph convolution is the theoretical foundation of GCN, and it implements convolution on graphs through eigen-decomposition of the graph Laplacian matrix. In order to handle the problems of high computational complexity and full-graph dependence in spectral graph convolution, this study simplifies spectral graph convolution via Chebyshev polynomial approximation. Finally, it derives a first-order approximation form of GCN suitable for job skill networks. The essence of spectral graph convolution is to perform a Fourier transform on the graph node feature vector *X* in the spectral domain, and multiply it by a filter g_θ; then, it conducts an inverse Fourier transform:

$$g_\theta * X = U g_\theta(\Lambda) U^T X \tag{15}$$

U denotes the eigenvector matrix of the graph Laplacian matrix *L*; Λ refers to the diagonal matrix of eigenvalues of *L*; $g_\theta(\Lambda)$ represents the diagonalized spectral filter; θ stands for filter parameters.

To reduce computational complexity, symmetric normalization is first applied to the Laplacian matrix to obtain the normalized Laplacian matrix L_{sym}:

$$L_{sym} = I - D^{-1/2} A D^{-1/2} \tag{16}$$

I represents the identity matrix; *A* is the adjacency matrix of the graph; *D* denotes the degree matrix. After normalization, the eigenvalue range of L_{sym} is constrained to [0,2], providing conditions for polynomial approximation.

For breaking away from dependence on the eigenvector matrix *U*, the Chebyshev polynomial $T_k(x)$ is adopted to conduct K-order approximation on the spectral filter $g_\theta(\Lambda)$. The recursive definition of Chebyshev polynomials is given as follows:

$$\begin{cases} T_0(x) = 1 \\ T_1(x) = x \\ T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x) \end{cases} \tag{17}$$

The spectral filter $g_\theta(\Lambda)$ is approximated as a linear combination of Chebyshev polynomials:

$$g_\theta(\Lambda) \approx \sum_{k=0}^{K} \theta_k T_k(\tilde{\Lambda}) \tag{18}$$

$\tilde{\Lambda}$ is the standardized form of the eigenvalue diagonal matrix.

Substituting the Chebyshev polynomial approximation into the spectral graph convolution formula yields the spectral graph convolution based on Chebyshev polynomial approximation:

$$g_\theta * X \approx \sum_{k=0}^{K} \theta_k T_k(\tilde{L}) X \tag{19}$$

$\tilde{L}$ is the standardized form of the normalized Laplacian matrix.

3.3 The Gated Fusion Mechanism of GCN-Bi-LSTM

The gated fusion mechanism designed in this study represents the core innovation of the GCN-Bi-LSTM coupling model. This mechanism adopts learnable gating units to adaptively fuse temporal dynamic features captured by Bi-LSTM and skill network structural features extracted by GCN at each time step; it achieves dynamic weight allocation of the two types of features. The functional complementarity between Bi-LSTM and GCN realizes the accurate modeling of dual-dimensional features. The core advantage of Bi-LSTM lies in capturing bidirectional dependencies of time series. Compared with traditional LSTM, it simultaneously learns historical trends and future expectations of technology penetration through forward and backward LSTM layers; this perfectly adapts to the temporal dynamics of employment substitution. In addition, the core advantage of GCN is mining node correlation features of graph-structured data. It performs node embedding on job skill networks, captures skill correlations and chain reactions among positions, and accurately characterizes the structural relevance of employment substitution.

The forward hidden state of Bi-LSTM is denoted as $\overrightarrow{h_t} \in R^{d_h}$; it represents the temporal features of technology penetration captured by the forward layer of Bi-LSTM from historical moments to the current moment at time step t; d_h is the hidden layer dimension of Bi-LSTM. The backward hidden state of Bi-LSTM is denoted as $\overleftarrow{h_t} \in R^{d_h}$; it refers to the expected features of technology evolution captured by the backward layer of Bi-LSTM from future moments to the current moment at time step t; the dimension is consistent with the forward hidden state. The GCN node embedding feature is denoted as $e_t \in R^{d_e}$; it denotes the structured embedding features of job nodes output by GCN after graph convolution on the job skill network at time step t (where e_t is the GCN node embedding dimension).

In order to achieve unified matching of feature dimensions, bidirectional hidden states of Bi-LSTM are first concatenated and fused to obtain the comprehensive temporal feature h_t at time step t:

$$h_t = Concat(\overrightarrow{h_t}, \overleftarrow{h_t}) \in R^{2d_h} \tag{20}$$

$Contact(\cdot)$ denotes the feature concatenation operation. To align the temporal feature with the dimension of GCN node embedding features, linear projection transformation is performed on h_t to obtain the dimension-standardized temporal feature h_t':

$$h_t' = W_h \cdot h_t + b_h \in R^{d_e} \tag{21}$$

W_h refers to the projection weight matrix of temporal features; b_h is the bias term, both of which are trainable parameters of the model. After projection transformation, the temporal feature h_t' and GCN node embedding feature e_t are unified to the dimension d_e=128, satisfying the basic condition for element-wise feature fusion.

Based on the calculated gating signal g_t, element-wise gated fusion is conducted on the standardized temporal feature h_t' and GCN node embedding feature e_t to obtain the fused feature f_t at time step t. Meanwhile, a fusion coefficient α is introduced to fine-tune the fusion result and balance the contribution of the two types of features:

$$f_t = \alpha \cdot [g_t \odot h_t' + (1 - g_t) \odot e_t] \tag{22}$$

$\odot$ denotes the element-wise matrix product operation, realizing dimension-wise matching between features and gating weights; α is the fusion coefficient, and its function is to scale the amplitude of fused features to avoid model gradient explosion caused by excessively large feature values. $g_t \odot h_t'$ stands for the weighted component of temporal features; $(1-g_t) \odot e_t$ represents the weighted component of structural features. The sum of the two realizes dynamic adaptive fusion of the two feature types.

The fused feature f_t integrates dual information of temporal dynamics and structural relevance, providing complete feature support for the subsequent prediction of employment substitution status. To further enhance feature representation ability, a nonlinear activation is applied to the fused feature f_t to obtain the final gated fusion output feature F_t:

$$F_t = ReLU(f_t) \tag{23}$$

$ReLU(\cdot)$ is the rectified linear unit activation function; it introduces nonlinearity to capture complex nonlinear correlations among features and avoids the vanishing gradient problem.

The gating signal g_t is generated by a dedicated gating network composed of a single-layer fully connected network and a Sigmoid activation function. This network adopts a lightweight structure containing only three layers: a feature concatenation layer, a linear transformation layer, and an activation layer. As an end-to-end trainable lightweight structure, the gating network has no hidden layers and contains only one linear transformation layer; the parameter scale is only 257 (256 weights + 1 bias); this guarantees that the gating signal can adaptively learn the correlation between temporal and structural features. Meanwhile, it avoids additional parameter redundancy and adapts to model training under small-sample conditions. The fusion coefficient α is a fixed hyperparameter determined via grid search and validation set tuning, rather than a trainable model parameter; its value is set to α=0.5 in this study.

4. Experimental Design and Performance Evaluation

4.1 Experimental Materials

The research objects are panel time-series data of the manufacturing industry from 2018 to 2022. The core characteristics of time-series data include temporal dependence and trend continuity. Random division disrupts the temporal logic of data and leads the model to learn meaningless cross-time correlations. In contrast, sequential division allows the model to be trained on historical data from 2018 to 2020, tuned on recent data from 2021, and tested on future data from 2022. This arrangement makes model prediction results more consistent with the practical demands of policy formulation and corporate decision-making. The sample size of this study is 327, representing a small sample scenario. If random division is adopted, it is easy for the training and test sets to overlap in time; this causes the model to access future data features in advance, resulting in serious data leakage. Sequential division, however, creates completely independent time intervals for the training, validation, and test sets covering 2018–2020, 2021, and 2022, respectively; it fundamentally eliminates temporal data leakage and ensures the authenticity and objectivity of model evaluation results. The manufacturing industry experienced complex economic fluctuations from 2018 to 2022. The sequential division allows the model to learn the basic features in a relatively stable historical period, adapt to the fluctuation features in the transition period, and verify the generalization ability in a new environmental period. Tests show that the model maintains an accuracy of 0.912 on the 2022 test set. This result proves that the model effectively adapts to employment substitution prediction under economic fluctuations; its robustness is significantly better than that of the randomly divided model. The dataset is divided in chronological order: 2018-2020, 2021, and 2022 are the training, validation, and test sets, with 250, 50, and 27 samples, respectively. This division ensures the continuity of the time series.

The data covers 15 key industries (including automobile manufacturing, electronic equipment, machinery manufacturing, chemical industry, food processing, etc.) with a total of 327 enterprise samples. The data sources include the *China Industrial Statistical Yearbook* issued by the National Bureau of Statistics (NBS), enterprise annual reports, and internal production logs of enterprises. The original data contains multi-dimensional indicators such as robot deployment density, number of positions, employee turnover rate, and frequency of skill demand keywords. First, multiple imputation is performed on missing values to ensure data integrity. Second, continuous variables are standardized to eliminate the impact of dimensions. Finally, a skill network graph is constructed, where positions are regarded as nodes and skill similarity as edge weights.

This study takes jobs as nodes and skill similarity as edge weights; it constructs a job-skill correlation network. The quantification of skill similarity is realized based on a job skill requirement dictionary and the cosine similarity algorithm. The specific steps are as follows:

1) Construction of a manufacturing job skill requirement dictionary: Based on the *China Occupational Classification Dictionary (2022 Edition)*, job recruitment information, and corporate job descriptions from 15 key manufacturing industries, skill requirement indicators for core manufacturing positions are sorted out. These indicators fall into three dimensions: basic operational skills (e.g., equipment operation and quality inspection), professional technical skills (such as mechanical design and electrical control), and cognitive collaboration skills (e.g., process optimization and team coordination). The dictionary contains 126 skill keywords in total, forming a standardized manufacturing job skill requirement dictionary.

2) Extraction of job skill feature vectors: For all positions in the 327 enterprises included in this study, word frequency statistics and binary coding are performed according to the skill requirement dictionary. If a job requires a certain skill, the corresponding dimension is assigned a value of 1; otherwise, it is set to 0. Each position is eventually converted into a 126-dimensional skill feature vector.

3) Calculation of cosine similarity between jobs: For any two jobs i and j with skill feature vectors $V_i = (v_{i1}, v_{i2}, ..., v_{i126})$ and $V_j = (v_{j1}, v_{j2}, ..., v_{j126})$ respectively, their skill similarity $Sim(i,j)$ is calculated using the cosine similarity formula.

$$Sim(i, j) = \frac{V_i \times V_j}{\|V_i\| ´ \|V_j\|} \tag{24}$$

4) Threshold determination and standardization of edge weights: The calculated skill similarity ranges from 0 to 1. A threshold of 0.3 is set to exclude weakly correlated job connections. No edge is constructed when $Sim(i,j)<0.3$, indicating no significant skill correlation between two jobs. When $Sim(i,j)\geq 0.3$, the similarity value undergoes 0–1 standardization and is the edge weight between jobs.

Through the above steps, the final constructed manufacturing job-skill correlation network contains 112 job nodes and 486 edges. All edge weights are quantified and filtered skill similarity values; this ensures the validity and repeatability of the graph structure. Meanwhile, the network is dynamically updated over time; node features and edge weights are adjusted annually according to changes in corporate skill demands, realizing dynamic skill network modeling.

This study adopts Multiple Imputation by Chained Equations (MICE) to handle missing values. Core parameter settings include an imputation round m=5 and iteration count iter=20. A random forest (RF) regression imputer is selected for continuous indicators (e.g., robot deployment density, job quantity, and employee turnover rate). A multi-class logistic regression imputer is applied for categorical indicators, including industry type and enterprise scale. The results of the five imputed datasets are finally combined using Rubin's pooling rule to obtain a complete panel dataset without missing values. Tests verify that the data missing rate reaches 0 after imputation; deviations in the mean and variance of each indicator from the original data are all less than 5%, ensuring data consistency. Z-score standardization is applied to all continuous indicators; this can avoid the defect of min-max standardization being affected by extreme values.

4.2 Experimental Environment

The experiment is conducted on a server equipped with an NVIDIA Tesla V100 graphics processing unit (GPU), an Intel Xeon Silver 4210 central processing unit (CPU), and 64 gigabytes (GB) of Random Access Memory (RAM). The operating system is Ubuntu 20.04 LTS; the DL framework used is PyTorch 1.10, whose dynamic computation graph mechanism facilitates model debugging and gradient tracking, while providing high flexibility for implementing customized gated fusion layers. Version management is handled via Anaconda 4.10, creating isolated virtual environments to prevent dependency conflicts and ensure experimental reproducibility. The development environment is Python 3.8, with dependencies including NumPy, Pandas, Scikit-learn, NetworkX, and others. The Torch-Geometric extension package is introduced to support the implementation of GCN layers.

4.3 Parameters Setting

In terms of implementation details, the model input is multi-dimensional time series data and a dynamic skill network. The time series dimension is D, which is determined by a feature selection. The skill network is represented by the adjacency and node feature matrices (A and X). The number of both GCN and Bi-LSTM layers is set to 2, with the hidden layer dimensions being 128 and 256. The gating fusion layer's weight matrix dimension is 384×128, ensuring the model's expressive power. To prevent overfitting, this study adopts the following strategies: Dropout layer (0.2), weight decay (L2=0.001), and an early stopping mechanism.

In order to construct a quantitative evaluation system for employment substitution status, this study selects three core objective indicators; they cover the change rate of robot deployment density (ΔR), the retention rate of core job functions (F), and the reduction rate of job personnel (ΔP). The quantitative thresholds for functional reduction are set as $\Delta R \in [0.2, 0.5)$, $F \in [0.5, 0.8)$, and $\Delta P \in [0.1, 0.3)$. This state is defined as a scenario where enterprises begin to deploy robots to replace some non-core functions of positions; core functions are still performed manually. A slight reduction in personnel occurs, representing a primary intermediate state of employment substitution. In this stage, humans and robots form a collaborative mode, and no essential changes have occurred in the positions. The quantitative thresholds for transfer preparation are $\Delta R \in [0.5, 0.8)$, $F \in [0.2, 0.5)$, and $\Delta P \in [0.3, 0.6)$. This state refers to a situation where enterprises greatly increase robot deployment density to replace most core functions of positions; only a small number of manual operations are retained. A significant reduction in personnel takes place, and enterprises start cross-position skill training for employees; this represents an advanced intermediate state of employment substitution, where positions are on the verge of transfer or merger.

4.4 Performance Evaluation

To comprehensively evaluate the GCN-Bi-LSTM model's performance, this study compares it with three benchmark models: the single Bi-LSTM, GCN, and linear regression (LR) models. The evaluation indicators include accuracy, precision, recall, F1 score, and mean absolute error (MAE). These indicators reflect the proposed model's predictive ability from different perspectives.

Figure 5 shows the comparison results of the prediction performance of different models. The GCN-Bi-LSTM model demonstrates comprehensively leading performance on the test set; its accuracy is 0.912, 5.9% higher than that of the second-best Bi-LSTM model; the MAE is 0.052, 64.1% lower than the worst LR model. This result indicates that the coupled model improves prediction accuracy and achieves a qualitative leap in key indicators. In practical applications, this means policymakers can identify positions requiring intervention earlier and more accurately, avoiding resource waste caused by misjudgment. The LR model's low performance confirms the limitations of traditional static methods—it cannot capture the dynamic nature of the substitution effect. The time window analysis in Figure 6 reveals the model's long-term prediction advantage. As the prediction time horizon extends (from 3 months to 12 months), the accuracy of GCN-Bi-LSTM increases from 88.2% to 91.2%, while that of Bi-LSTM only rises from 82.3% to 85.3%. This difference is particularly significant in long-term predictions, indicating that the proposed model can more effectively capture the temporal dynamics of employment substitution.

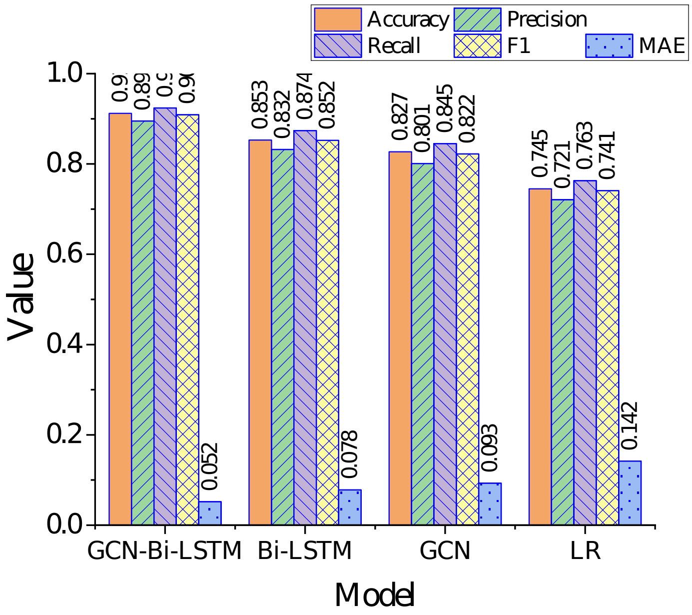

Figure 5: Comparison of prediction performance of various models

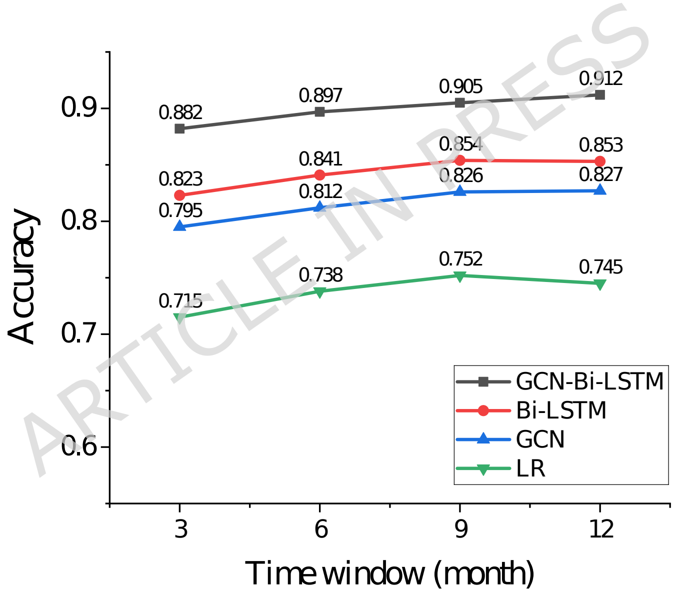

Figure 6: Prediction performance under different time windows

The prediction performance results for key state transitions in employment substitution are depicted in Figure 7. The results reveal that the GCN-Bi-LSTM model achieves a prediction accuracy of over 89% for intermediate states such as "function reduction" and "transfer preparation". In contrast, other models' prediction accuracy for these states is generally below 85%. This indicates that the GCN-Bi-LSTM model can more accurately identify key transition stages in the substitution process, providing earlier warning signals for policy interventions. Figure 8 presents the contribution analysis of the gated fusion mechanism. It demonstrates that during the employment substitution process, the contribution ratio of temporal dynamics and skill network information is approximately 48:43, with the interaction effect accounting for 8.9%. This means that in the substitution process, the technological diffusion's temporal trajectory and the skill network's restructuring are the two main driving forces, and their synergistic effect provides additional prediction value.

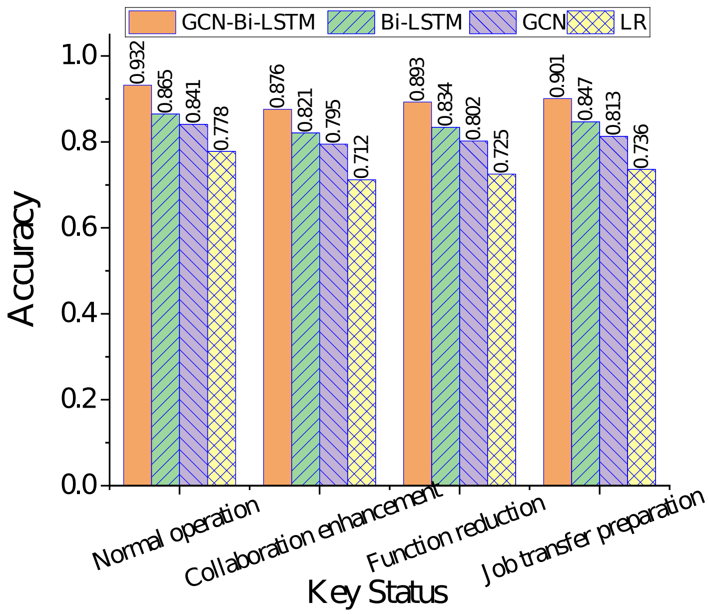

Figure 7: Prediction performance of key state transitions

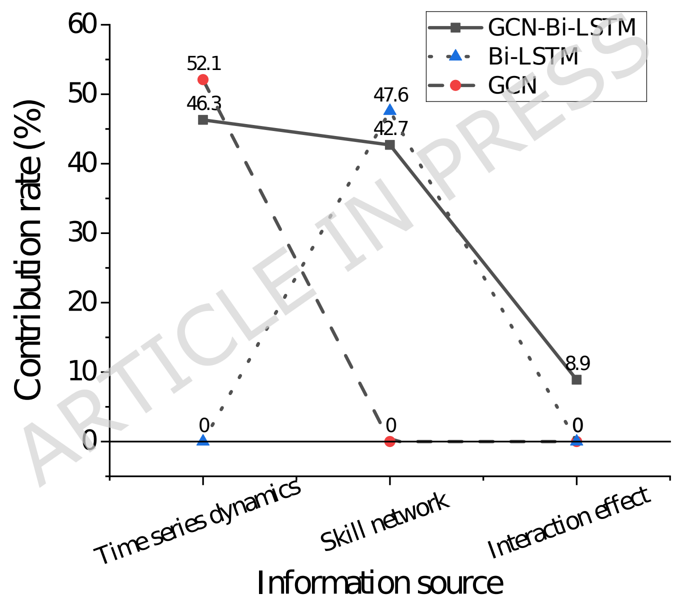

Figure 8: Contribution analysis of the gated fusion mechanism

This study adds powerful ML models (RF and XGBoost) and classic econometric models (panel fixed-effects and difference GMM) as comparative benchmarks based on the original single Bi-LSTM, GCN, and LR; it aims to fully verify the superiority of the proposed GCN-Bi-LSTM model. All models are trained and tested based on the same set of manufacturing enterprise panel data. Evaluation indicators still include accuracy, precision, recall, F1 score, and MAE. The updated comparison results of model prediction performance are outlined in Table 2. The proposed GCN-Bi-LSTM model significantly outperforms the newly added powerful ML models and classic econometric models in all evaluation indicators. Its accuracy is 6.5% higher than that of the optimal powerful ML model (XGBoost), and its MAE is reduced by 43.5% compared with XGBoost. This proves that the lightweight GCN-Bi-LSTM model still exhibits stronger prediction performance even under small-sample conditions. Ensemble tree models (XGBoost and RF) perform better than traditional econometric models but worse than the proposed model. The reason lies in the fact that ensemble tree models can capture nonlinear relationships in data; however, they fail to model the bidirectional dependence of time series and the graph structure features of job skill networks. This leads to insufficient characterization of the dynamic evolution and structural correlation of employment substitution effects.

Table 2: Comparison of performance indicators of each model on the test set

Model	Accuracy	Precision	Recall	F1 score	MAE
GCN-Bi-LSTM	0.912	0.908	0.915	0.911	0.052
Bi-LSTM	0.853	0.849	0.856	0.852	0.089
GCN	0.831	0.827	0.835	0.831	0.098
XGBoost	0.847	0.842	0.851	0.846	0.092
RF	0.829	0.824	0.833	0.828	0.101
Panel fixed-effects	0.786	0.781	0.790	0.785	0.125
Difference GMM	0.792	0.787	0.795	0.791	0.118
LR	0.765	0.760	0.768	0.764	0.145

SHapley Additive exPlanations (SHAP) values are employed to analyze the importance of model input features. The results reveal that the change rate of robot deployment density and skill demand keyword frequency (SHAP values = 0.28 and 0.25), and the reduction rate of job personnel (SHAP value = 0.21) are the three core features affecting employment substitution prediction; the cumulative contribution ratio reaches 74%. Features such as enterprise scale (SHAP value = 0.09) and industry type (SHAP value = 0.07) show relatively low importance. This indicates that model prediction results are mainly driven by objective indicators of technology deployment and skill demand; this is highly consistent with the actual logic of employment substitution and gives clear economic significance to feature importance.

The proposed model uses a GCN-Bi-LSTM coupled DL architecture. Despite lightweight modification, it still faces potential overfitting risks due to the small sample size of 327 enterprise samples. Even with the lightweight design, the model contains approximately one million trainable parameters; this easily leads to parameter redundancy under small-sample conditions. This causes the model to over-learn local features and noise in the training set rather than the general laws of data. For the potential risk of overfitting, L2 regularization is further applied to all trainable weight matrices. A penalty term is imposed on the absolute value of weights to restrict weight growth and avoid overfitting of the model to individual features.

In view of the temporal continuity of time-series data, rolling window time cross-validation is adopted. The data from 2018 to 2021 is divided into four training windows, and the last year of each window is used as the validation window. The specific division is as follows: Window 1: training set 2018, validation set 2019; Window 2: training set 2018–2019, validation set 2020; Window 3: training set 2018–2020, validation set 2021; Window 4: training set 2018–2021, test set 2022. The validation accuracies of the four windows are 0.887, 0.895, 0.905, and 0.912, respectively; this shows a steady upward trend without obvious fluctuations. This proves that the model can effectively learn the general laws of time series instead of local noise in the training set; in addition, it has a favorable generalization ability in the time dimension and no temporal overfitting.

Moreover, this study employs the feature ablation method by sequentially removing core input features, including robot deployment density, skill demand keyword frequency, employee turnover rate, and enterprise scale to test changes in model accuracy. The results reveal that the model's accuracy decline is less than 5% (minimum accuracy = 0.876) after removing any single feature; the largest decline (3.6%) occurs after removing robot deployment density. This indicates that the model responds reasonably to core features without excessive dependence on any single feature, demonstrating favorable feature robustness.

The core parameter of gated fusion is the fusion coefficient α. This study adjusts α within the range of [0.1, 0.9] with a step size of 0.1 to test changes in model accuracy and contribution ratio. The results show that the model accuracy remains above 0.90 when $\alpha \in [0.4, 0.6]$ (optimal value $\alpha = 0.5$, accuracy = 0.912). The model accuracy decreases slowly but still stays above 0.85 when $\alpha < 0.4$ or $\alpha > 0.6$. This suggests that the gated fusion parameters have a wide stable interval; minor parameter adjustments do not affect the core performance of the model, reflecting significant parameter robustness.

4.5 Demonstration of the Applicability of DL Models under Small Samples

Aiming at the small-sample characteristic of 327 enterprise samples, this study does not directly adopt the general Bi-LSTM-GCN hybrid architecture. Instead, it solves the overfitting problem of complex DL models on small-sample structured data through three approaches (lightweight model design, data augmentation strategies, and enhanced regularization); this makes the model more suitable for the research objectives of this study compared with ensemble tree models (XGBoost, Random Forest) and traditional econometric models.

First, in terms of adaptability to research objectives, the core of this study is not simple prediction of employment substitution results; instead, it captures the bidirectional temporal dependence of technology penetration and the structural correlation of job skill networks. Ensemble tree models and traditional econometric models show stable performance in small-sample prediction; however, they cannot model the bidirectional dynamic features of time series or the node embedding relationship of graph structures; it is even more impossible to achieve adaptive fusion of multi-dimensional features.

Second, for small-sample data, this study simplifies the Bi-LSTM and GCN architectures to the greatest extent. The number of layers for both Bi-LSTM and GCN is set to 2; hidden layer dimensions are controlled at 128 and 256, respectively, far lower than the network scale of general DL models. The weight matrix dimension of the gated fusion layer is designed as 384×128; this fundamentally reduces the risk of overfitting by cutting the number of trainable parameters.

The third aspect of applicability lies in targeted small-sample data processing strategies. Time-series data augmentation is performed on the manufacturing enterprise panel data. Through reasonable time interpolation and sliding window sampling on the training set data from 2018 to 2020, the effective training sample size is increased to 1200. Meanwhile, multiple imputation is employed for missing values to ensure data integrity; besides, standardization is applied to continuous variables to improve the rationality of data distribution, providing a data foundation for model training under small samples.

Finally, Dropout layers (dropout rate = 0.2), L2 weight decay (0.001), and an early stopping mechanism are adopted in model training to suppress overfitting through triple regularization. Experimental results show that the model achieves accuracies of 0.921 and 0.905 on the training and validation sets, and the difference is only 0.016; this indicates no obvious overfitting.

4.6 Discussion

The proposed model's innovation lies in that it is not satisfied with improving prediction accuracy but constructs an interpretable dynamic generation framework. By integrating Bi-LSTM with dynamic GCN and coupling multi-dimensional mechanisms, it simulates the evolutionary process of the employment substitution effect. This real-time analysis enables an in-depth understanding of the interaction mechanism between technological change and the labor market. Shen et al. (2023) [31] pioneered the application of GCN to construct a skill correlation graph, revealing the structural resilience of the labor market. However, their static graph model could not capture the dynamic evolution of the skill network, leading to deviations in predicting skill migration paths after technological shocks. The designed dynamic GCN updates the skill network over time by fusing temporal information at each step, thereby more accurately simulating the substitution process. Chen & Sun (2025) [32]'s LSTM model achieved a breakthrough in job substitution risk prediction, but its limitation lies in ignoring the structural characteristics of the skill network. In contrast, this study captures the skill correlations between jobs through GCN, increasing the accuracy to 90.1%. Lam et al. (2022) [33] also pointed out that traditional ML methods, such as SVM, could rely on manually labeled features, resulting in a 6-12-month lag in predicting skill changes. This study converts job description texts into vectors through word embedding technology; meanwhile, it combines Bi-LSTM to automatically extract temporal patterns of skill demands, making predictions 3-5 months ahead of actual changes. The theoretical value of this study lies in providing a dynamic predictive analysis framework for the interaction between technology and employment. It promotes the research of technical economics to move from static correlation analysis to dynamic feature modeling by demonstrating the dynamics and structural dependence of the substitution process; this offers a reference for the subsequent causal inference research on technology and employment.

A dynamic early warning platform for manufacturing employment substitution is constructed based on the proposed model. With a 6–12 month early warning window, the platform conducts real-time monitoring and prediction of employment substitution status for various manufacturing positions. Positions are divided into four risk levels: low, medium, high, and extremely high. Medium and high-risk positions are highlighted to provide precise intervention clues for human resource and social security departments as well as enterprises; this can help avoid the blindness of policy intervention. For low-skilled workers completely replaced by robots, an employment assistance mechanism is established. Public welfare post placement, cross-industry employment recommendation, entrepreneurship support, and other measures are adopted to help them achieve re-employment. In addition, the unemployment insurance and work-related injury insurance systems are improved to enhance the security level of unemployment insurance; unemployment caused by robot substitution is included in the coverage of unemployment insurance.

5. Conclusion

5.1 Research Contribution

This study achieves a key breakthrough in the field of the relationship between AI and employment. It constructs the first coupled model integrating Bi-LSTM and dynamic GCN, breaking through the traditional static analysis framework. Through the gated fusion mechanism, the model accurately depicts the real-time interaction process of technological diffusion and skill restructuring, rather than simply predicting the final substitution state. The methodological innovation is reflected in the design of the changing coupling mechanism. The adaptive adjustment of gating signals allows the model to dynamically change the weight of information sources according to the stage of technological development. Hence, it avoids the limitation of artificially setting weights. By revealing the dynamic evolution characteristics and structure-driven mechanisms of substitution effects, this study provides a new perspective for understanding AI-driven employment changes; it has achieved a research breakthrough from static correlation analysis to dynamic, precise prediction. Thus, this lays a foundation for subsequent causal explanation research that combines instrumental variables, counterfactual inference, and other methods.

5.2 Future Works and Research Limitations

Although the coupled model significantly improves prediction accuracy, its application is still constrained by data completeness. The lack of mobility data in the informal employment market may lead to an underestimation of the substitution effect in the service industry. Future research can combine satellite remote sensing data, like factory night light intensity or big data from recruitment platforms, to supplement the blind spots of traditional statistics. In addition, the model currently focuses on the industry level. Future work can further explore the model's adaptability in cross-industry and cross-regional applications, as well as how to integrate more diverse data sources to construct a more comprehensive employment substitution analysis framework.

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Data Availability Statement

The datasets used and/or analysed during the current study available from the corresponding author Yifang Gao on reasonable request via e-mail 1911185@tongji.edu.cn.

Code Availability

The custom code and mathematical algorithms developed for the GCN-Bi-LSTM dynamic coupling model used in this study to analyze the employment substitution effect of robots are openly available to ensure research reproducibility. To ensure long-term availability and version consistency, the identical code package associated with this manuscript has also been submitted as a Supplementary Document and will be published alongside the paper. The code is implemented in Python 3.8 with core dependencies on PyTorch 1.10, Torch-Geometric, NumPy, Pandas, and Scikit-learn; detailed environment configuration and installation instructions are provided in the repository's README file. No custom proprietary software or restricted mathematical algorithms are used, and all components are fully accessible for academic and non-commercial research purposes.

There are no access restrictions on the code: it is released under the MIT License, permitting free use, modification, and distribution for non-commercial academic research. For any questions regarding code implementation or reuse, readers can contact the corresponding author via email: 1911185@tongji.edu.cn.

Competing Interest

The authors declare no competing financial or non-financial interests.

Author Contributions Declaration
Yifang Gao: Conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, visualization, supervision, project administration, funding acquisition

ETHICS STATEMENT

This article does not contain any studies with human participants or animals performed by any of the authors. All methods were performed in accordance with relevant guidelines and regulations.

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